

## CALDWELL, ID, USA

WMO: 726813

Lat: 43.650N

Lon: 116.633W

Elev: 740

StdP: 92.74

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94195

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.5      | -9.1 | -17.2                         | 0.9 | -11.3 | -13.9 | 1.2 | -8.5 | 11.4                  | 3.3  | 10.0 | 3.2  | 1.3                   | 120  | 0.548 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.1                   | 36.3            | 19.0 | 34.1 | 18.3 | 32.8 | 17.8 | 20.2                | 33.7 | 19.3 | 32.5 | 18.5 | 31.4 | 2.5                  | 350  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 15.2                            | 11.8 | 25.4 | 13.9 | 10.8 | 25.1 | 12.8 | 10.1 | 24.3 | 61.0          | 33.7 | 57.7 | 32.8 | 55.0 | 31.3 | 25.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 9.9           | 8.6  | 7.6 | DB                         | -14.6 | 38.8               | 4.6 | 1.7                                                | -18.0 | 40.1       | -20.7 | 41.1       | -23.3 | 42.0       | -26.6 | 43.3 | (4) |
| (5)               |      |     | WB                         | -15.1 | 22.4               | 4.4 | 1.2                                                | -18.3 | 23.2       | -20.9 | 23.9       | -23.3 | 24.6       | -26.5 | 25.4 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.9  | -0.6  | 2.6   | 6.9   | 10.0  | 14.8  | 19.2  | 24.1  | 22.6  | 17.4  | 10.2  | 4.1   | -0.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 9.15  | 4.41  | 3.80  | 3.47  | 3.48  | 3.93  | 3.83  | 2.81  | 2.90  | 3.60  | 3.74  | 4.12  | 4.47 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1242  | 327   | 209   | 107   | 43    | 6     | 0     | 0     | 1     | 42    | 179   | 328   |      | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3123  | 586   | 442   | 354   | 251   | 123   | 36    | 2     | 4     | 59    | 254   | 426   | 586  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1583  | 0     | 1     | 11    | 42    | 153   | 275   | 436   | 392   | 223   | 47    | 3     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 422   | 0     | 0     | 0     | 0     | 12    | 61    | 179   | 137   | 32    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 6328  | 0     | 0     | 1     | 31    | 260   | 922   | 2524  | 1959  | 602   | 30    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 3091  | 0     | 0     | 0     | 3     | 77    | 387   | 1393  | 1003  | 222   | 4     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.9   | 2.6   | 3.3   | 3.6   | 3.7   | 3.3   | 3.1   | 2.6   | 2.5   | 2.5   | 2.5   | 2.8   | 2.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 280   | 37    | 25    | 29    | 26    | 23    | 20    | 7     | 10    | 15    | 19    | 33    | 35   | (15) |
| (16) |                                                                                  | PrecMax     | 441   | 126   | 86    | 87    | 83    | 65    | 66    | 31    | 58    | 64    | 40    | 67    | 102  | (16) |
| (17) |                                                                                  | PrecMin     | 147   | 7     | 0     | 2     | 0     | 2     | 0     | 0     | 0     | 0     | 0     | 5     | 2    | (17) |
| (18) |                                                                                  | PrecStd     | 69    | 25    | 18    | 20    | 17    | 16    | 16    | 9     | 15    | 17    | 12    | 18    | 23   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 11.0  | 16.0  | 21.8  | 27.1  | 32.2  | 36.3  | 38.6  | 37.4  | 33.9  | 27.2  | 17.8  | 12.3 | (19) |
| (20) |                                                                                  |             | MCWB  | 6.9   | 9.1   | 11.0  | 14.3  | 17.1  | 19.3  | 20.2  | 18.8  | 17.9  | 14.4  | 10.1  | 7.8  | (20) |
| (21) |                                                                                  | 2%          | DB    | 7.9   | 12.7  | 18.2  | 23.0  | 28.8  | 32.9  | 37.1  | 35.5  | 31.6  | 22.9  | 14.9  | 9.1  | (21) |
| (22) |                                                                                  |             | MCWB  | 4.8   | 7.0   | 9.4   | 12.4  | 15.6  | 17.8  | 19.4  | 18.6  | 16.9  | 12.7  | 8.9   | 5.9  | (22) |
| (23) |                                                                                  | 5%          | DB    | 6.7   | 10.9  | 16.3  | 21.0  | 26.3  | 31.1  | 35.4  | 33.8  | 29.0  | 20.5  | 12.7  | 7.5  | (23) |
| (24) |                                                                                  |             | MCWB  | 3.8   | 6.2   | 8.5   | 11.2  | 14.4  | 17.2  | 19.0  | 17.9  | 15.9  | 11.6  | 7.6   | 4.6  | (24) |
| (25) |                                                                                  | 10%         | DB    | 5.1   | 8.6   | 13.9  | 17.8  | 23.7  | 28.6  | 33.5  | 32.4  | 27.2  | 17.8  | 11.0  | 5.8  | (25) |
| (26) |                                                                                  |             | MCWB  | 2.4   | 5.0   | 7.3   | 9.6   | 13.5  | 16.1  | 18.3  | 17.3  | 15.0  | 10.1  | 6.6   | 3.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 7.3   | 9.5   | 11.8  | 14.7  | 18.0  | 20.7  | 21.9  | 20.2  | 19.4  | 15.1  | 11.3  | 8.3  | (27) |
| (28) |                                                                                  |             | MCDWB | 9.6   | 14.9  | 19.1  | 25.5  | 29.6  | 34.4  | 35.4  | 34.1  | 30.5  | 25.5  | 15.4  | 11.6 | (28) |
| (29) |                                                                                  | 2%          | WB    | 5.4   | 7.6   | 10.3  | 12.9  | 16.3  | 19.0  | 20.7  | 19.4  | 17.9  | 13.4  | 9.7   | 6.5  | (29) |
| (30) |                                                                                  |             | MCDWB | 7.7   | 12.0  | 16.9  | 21.7  | 26.9  | 31.1  | 34.4  | 33.2  | 28.7  | 21.8  | 14.3  | 9.0  | (30) |
| (31) |                                                                                  | 5%          | WB    | 4.0   | 6.4   | 9.2   | 11.5  | 15.1  | 17.7  | 19.8  | 18.6  | 16.8  | 12.0  | 8.0   | 4.9  | (31) |
| (32) |                                                                                  |             | MCDWB | 6.3   | 10.0  | 15.4  | 19.2  | 25.0  | 29.3  | 33.4  | 32.3  | 27.2  | 19.1  | 12.2  | 7.3  | (32) |
| (33) |                                                                                  | 10%         | WB    | 2.8   | 5.1   | 7.9   | 10.3  | 14.0  | 16.6  | 18.9  | 17.8  | 15.5  | 10.6  | 6.7   | 3.3  | (33) |
| (34) |                                                                                  |             | MCDWB | 5.0   | 8.2   | 13.7  | 17.3  | 22.8  | 27.6  | 32.3  | 30.9  | 25.8  | 17.2  | 10.3  | 5.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 6.9   | 9.5   | 12.3  | 13.3  | 14.0  | 15.5  | 18.1  | 18.3  | 16.9  | 14.2  | 10.2  | 7.1   | (35) |      |
| (36) |                                                                                  | 5% DB       | MCDBR | 8.3   | 12.0  | 15.8  | 18.7  | 18.4  | 19.8  | 20.6  | 20.2  | 17.7  | 12.7  | 8.4   | (36) |      |
| (37) |                                                                                  |             | MCWBR | 6.2   | 7.8   | 9.2   | 9.9   | 8.5   | 8.5   | 8.2   | 7.9   | 8.7   | 9.6   | 8.1   | 6.1  | (37) |
| (38) |                                                                                  |             | MCDBR | 7.8   | 11.0  | 14.3  | 16.8  | 16.9  | 18.7  | 19.3  | 19.2  | 17.9  | 16.0  | 11.7  | 7.6  | (38) |
| (39) |                                                                                  | 5% WB       | MCWBR | 6.0   | 7.3   | 8.5   | 9.1   | 7.9   | 8.2   | 8.1   | 7.6   | 7.6   | 8.8   | 7.7   | 5.7  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.265 | 0.273 | 0.288 | 0.311 | 0.331 | 0.329 | 0.336 | 0.339 | 0.316 | 0.290 | 0.273 | 0.266 | (40) |      |
| (41) |                                                                                  | taud        | 2.574 | 2.592 | 2.563 | 2.485 | 2.446 | 2.484 | 2.470 | 2.442 | 2.508 | 2.578 | 2.579 | 2.564 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 893   | 943   | 962   | 956   | 939   | 939   | 928   | 915   | 916   | 905   | 873   | 856   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 64    | 75    | 88    | 103   | 110   | 107   | 107   | 106   | 92    | 75    | 63    | 59    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.56  | 2.58  | 3.89  | 5.46  | 6.63  | 7.54  | 7.74  | 6.68  | 5.18  | 3.39  | 1.97  | 1.29  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.23  | 0.34  | 0.38  | 0.43  | 0.46  | 0.39  | 0.27  | 0.28  | 0.28  | 0.25  | 0.17  | 0.19  | (45) |      |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (11 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page